Chapter 1: Red blood cell function

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Where I left off in the last video, we talked about how the hemoglobin in red blood cells is what sops up all of the

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oxygen so that it increases the diffusion gradient-- or it increases the incentive, we could say, for the oxygen to go across the membrane.

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We know that the oxygen molecules don't know that there's less oxygen here, but if you watch the video on diffusion you know how that process happens.

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If there's less concentration here than there, the oxygen will diffuse across the membrane and there's less inside the plasma because the hemoglobin is sucking it all up like a sponge.

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Now, one interesting question is, why does the hemoglobin even have to reside within the red blood cells?

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Why aren't hemoglobin proteins just freely floating in the blood plasma? That seems more efficient.

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You don't have to have things crossing through, in and out of, these red blood cell membranes. You wouldn't have to make red blood cells.

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What's the use of having these containers of hemoglobin? It's actually a very interesting idea.

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If you had all of the hemoglobin sitting in your blood plasma, it would actually hurt the flow of the blood.

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The blood would become more viscous or more thick.

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I don't want to say like syrup, but it would become thicker than blood is right now-- and by packaging the hemoglobin inside these containers, inside the red

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blood cells, what it allows the blood to do is flow a lot better. Imagine if you wanted to put syrup in water.

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If you just put syrup straight into water, what's going to happen?

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The water's going to become a little syrupy, a little bit more viscous and not flow as well.

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So what's the solution if you wanted to transport syrup in water?

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Well, you could put the syrup inside little containers or inside little beads and then let the beads flow in the water and then the water wouldn't be all gooey-- and that's exactly what's happening inside of our blood.

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Instead of having the hemoglobin sit in the plasma and make it gooey, it sits inside these beads that we

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1 minute, 58 seconds

call red blood cells that allows the flow to still be non-viscous.

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2 minutes, 4 seconds

So I've been all zoomed in here on the alveolus and these capillaries, these pulmonary capillaries-- let's zoom out a

Chapter 2: Pulmonary circulation

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2 minutes, 12 seconds

little bit-- or zoom out a lot-- just to understand, how is the blood flowing?

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2 minutes, 17 seconds

And get a better understanding of pulmonary arteries and veins relative to the other arteries and veins that are in the body.

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2 minutes, 25 seconds

So here-- I copied this from Wikipedia, this diagram of the human circulatory system-- and here in the back you can see the lungs.

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2 minutes, 34 seconds

Let me do it in a nice dark color. So we have our lungs here.

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2 minutes, 44 seconds

You can see the heart is sitting right in the middle.

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2 minutes, 46 seconds

And what we learned in the last few videos is that we have our little alveoli and our lungs.

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2 minutes, 51 seconds

Remember, we get to them from our bronchioles, which are branching off of the bronchi, which branch off of the trachea, which connects to our larynx, which connects to our pharynx, which connects to our mouth and nose.

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3 minutes, 1 second

But anyway, we have our little alveoli right there and then we have the capillaries.

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So when we go away from the heart-- and we're going to delve a little bit into the heart in this video as well--

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3 minutes, 17 seconds

so when blood travels away from the heart, it's de-oxygenated. It's this blue color. So this right here is blood.

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3 minutes, 24 seconds

This right here is blood traveling away from the heart. It's going behind these two tubes right there. So this is the blood going away from the heart.

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3 minutes, 33 seconds

So this blue that I've been highlighting just now, these are the pulmonary arteries and then they keep splitting into

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3 minutes, 43 seconds

arterials and all of that and eventually we're in capillaries-- super, super small tubes.

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3 minutes, 48 seconds

They run right past the alveoli and then they become oxygenated and now we're going back to the heart.

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3 minutes, 58 seconds

So we're talking about pulmonary veins. So we go back to the heart. So these capillaries-- in the capillaries we get oxygen.

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4 minutes, 5 seconds

Now we're going to go back to the heart. Hope you can see what I'm doing. And we're going to enter the heart on this side.

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4 minutes, 15 seconds

You actually can't even see where we're entering the heart.

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4 minutes, 17 seconds

We're going to enter the heart right over here-- and I'm going to go into more detail on that. Now we have oxygenated blood. It's red. And then that gets pumped out to the rest of the body.

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4 minutes, 26 seconds

Now this is the interesting thing.

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4 minutes, 30 seconds

When we're talking about pulmonary arteries and veins-- remember, the pulmonary artery was blue.

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4 minutes, 35 seconds

As we go away from the heart, we have de-oxygenated blood, but it's still an artery.

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4 minutes, 40 seconds

Then as we go towards the heart from the lungs, we have a vein, but it's oxygenated.

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5 minutes, 12 seconds

So that's this little loop here that we start and I'm going to keep going over the circulation pattern because the heart can get a little confusing, especially because of its three-dimensional nature.

Chapter 3: Heart anatomy and flow

5:22

5 minutes, 22 seconds

But what we have is, the heart pumps de-oxygenated blood from the right ventricle. You're saying, hey, why is it the right ventricle?

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5 minutes, 29 seconds

That looks like the left side of the drawing, but it's this dude's right-hand side, right? This is this guy's right hand.

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5 minutes, 37 seconds

And this is this dude's left hand. He's looking at us, right? We don't care about our right or left. We care about this guy's right and left. And he's looking at us.

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5 minutes, 44 seconds

He's got some eyeballs and he's looking at us. So this is his right ventricle. Actually, let me just start off with the whole cycle.

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5 minutes, 51 seconds

So we have de-oxygenated blood coming from the rest of the body, right?

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6 minutes

The name for this big pipe is called the inferior vena cava-- inferior because it's coming up below.

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6 minutes, 7 seconds

Actually, you have blood coming up from the arms and the head up here.

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6 minutes, 11 seconds

They're both meeting right here, in the right atrium.

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6 minutes, 19 seconds

Let me label that. I'm going to do a big diagram of the heart in a second. And why are they de-oxygenated?

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6 minutes, 27 seconds

Because this is blood returning from our legs if we're running, or returning from our brain, that had to

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6 minutes, 35 seconds

use respiration-- or maybe we're working out and it's returning from our biceps, but it's de-oxygenated blood. It shows up right here in the right atrium.

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6 minutes, 42 seconds

It's on our left, but this guy's right-hand side.

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6 minutes, 45 seconds

From the right atrium, it gets pumped into the right ventricle. It actually passively flows into the right ventricle.

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6 minutes, 53 seconds

The ventricles do all the pumping, then the ventricle contracts and pumps this blood right here-- and you don't see it, but it's going behind this part right here.

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7 minutes, 2 seconds

It goes from here through this pipe. So you don't see it.

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7 minutes, 6 seconds

I'm going to do a detailed diagram in a second-- into the pulmonary artery. We're going away from the heart. This was a vein, right?

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7 minutes, 14 seconds

This is a vein going to the heart. This is a vein, inferior vena cava vein.

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7 minutes, 24 seconds

This is superior vena cava. These are veins. They're de-oxygenated.

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7 minutes, 27 seconds

Then I'm pumping this de-oxygenated blood away from the heart to the lungs. Now this de-oxygenated blood, this is in an artery, right?

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7 minutes, 35 seconds

This is in the pulmonary artery. It gets oxygenated and now it's a pulmonary vein.

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7 minutes, 41 seconds

And once it's oxygenated, it shows up here in the left-- let me do a better color than that-- it shows up right here in the left atrium.

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7 minutes, 50 seconds

Atrium, you can imagine-- it's kind of a room with a skylight or that's open to the outside and in both of these cases,

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7 minutes, 57 seconds

things are entering from above-- not sunlight, but blood is entering from above. On the right atrium, the blood is entering from above.

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8 minutes, 5 seconds

And in the left atrium, the blood is entering-- and remember, the left atrium is on the right-hand side from our point of view-- on the left atrium, the blood is

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8 minutes, 12 seconds

entering from above from the lungs, from the pulmonary veins. Veins go to the heart.

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8 minutes, 20 seconds

Then it goes into-- and I'll go into more detail-- into the left ventricle and then the left ventricle pumps that

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8 minutes, 27 seconds

oxygenated blood to the rest of the body via the non-pulmonary arteries. So everything pumps out.

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8 minutes, 36 seconds

Let me make it a nice dark, non-blue color. So it pumps it out through there. You don't see it right here, the way it's drawn. It's a little bit of a strange drawing.

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8 minutes, 45 seconds

It's hard to visualize, but I'll show it in more detail and then it goes to the rest of the body. Let me show you that detail right now.

Chapter 4: Systemic circulation path

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8 minutes, 54 seconds

So we said, we have de-oxygenated blood. Let's label it right here. This is the superior vena cava.

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9 minutes, 1 second

This is a vein from the upper part of our body from our arms and heads. This is the inferior vena vaca.

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9 minutes, 6 seconds

This is veins from our abdomen and from our legs and the rest of our body. So it it first enters the right atrium.

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9 minutes, 14 seconds

Remember, we call the right atrium because this is someone's heart facing us, even though this is on the left-hand side. It enters through here.

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9 minutes, 22 seconds

It's de-oxygenated blood. It's coming from veins. the body used the oxygen. Then it shows up in the right ventricle, right?

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9 minutes, 31 seconds

These are valves in our heart.

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9 minutes, 34 seconds

And it passively, once the right ventricle pumps and then releases, it has a vacuum and it pulls more blood from the right atrium. It pumps again and then it pushes it through here.

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9 minutes, 44 seconds

Now this blood right here-- remember, this one still is de-oxygenated blood.

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9 minutes, 49 seconds

De-oxygenated blood goes to the lungs to become oxygenated.

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9 minutes, 54 seconds

So this right here is the pulmonary-- I'm using the word pulmonary because it's going to or from the lungs.

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10 minutes, 2 seconds

It's dealing with the lungs. And it's going away from the heart. It's the pulmonary artery and it is de-oxygenated.

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10 minutes, 17 seconds

Then it goes to the heart, rubs up against some alveoli and then gets oxygenated and then it comes right back.

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10 minutes, 31 seconds

Now this right here, we're going to the heart. So that's a vein.

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10 minutes, 40 seconds

It's in the loop with the lungs so it's a pulmonary vein and it rubbed up against the alveoli and got the oxygen diffused into it so it is oxygenated.

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10 minutes, 53 seconds

And then it flows into your left atrium.

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10 minutes, 56 seconds

Now, the left atrium, once again, from our point of view, is on the right-hand side, but from the dude looking at it, it's his left-hand side. So it goes into the left atrium.

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11 minutes, 4 seconds

Now in the left ventricle, after it's done pumping, it expands and that oxygenated blood flows into the left ventricle.

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11 minutes, 11 seconds

Then the left ventricle-- the ventricles are what do all the pumping-- it squeezes and then it pumps the blood into the aorta.

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11 minutes, 22 seconds

This is an artery. Why is it an artery? Because we're going away from the heart.

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11 minutes, 30 seconds

Is it a pulmonary artery? No, we're not dealing with the lungs anymore.

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11 minutes, 35 seconds

We dealt with the lungs when we went from the right ventricle, went to the lungs in a loop, back to the left atrium. Now we're in the left ventricle.

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11 minutes, 43 seconds

We pump into the aorta. Now this is to go to the rest of the body.

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11 minutes, 46 seconds

This is an artery, a non-pulmonary artery-- and it is oxygenated.

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11 minutes, 53 seconds

So when we're dealing with non-pulmonary arteries, we're oxygenated, but a pulmonary artery has no oxygen. It's going away from the heart to get the oxygen.

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12 minutes, 2 seconds

Pulmonary vein comes from the lungs to the heart with oxygen, but the rest of the veins go to the heart without oxygen because they want to go into that loop on the pulmonary loop right there.

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12 minutes, 13 seconds

So I'll leave you there.

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12 minutes, 14 seconds

Hopefully that gives-- actually, let's go back to that first diagram.

Chapter 5: Circulatory system overview

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12 minutes, 17 seconds

I think you have a sense of how the heart is dealing, but let's go look at the rest of the body and just get a sense of things.

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12 minutes, 25 seconds

You can look this up on Wikipedia if you like.

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12 minutes, 27 seconds

All of these different branching points have different names to them, but you can see right here you have kind of a branching off, a little bit below the heart.

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12 minutes, 37 seconds

This is actually the celiac trunk. Celiac, if I remember correctly, kind of refers to an abdomen.

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So this blood that-- your hepatic artery. Hepatic deals with the liver.

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12 minutes, 53 seconds

Your hepatic artery branches off of this to get blood flow to the liver.

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It also gives blood flow to your stomach so it's very important in digestion and all that. And then let's say this is the hepatic trunk.

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13 minutes, 5 seconds

Your liver is sitting like that. Hepatic trunk-- it delivers oxygen to the liver. The liver is doing respiration.

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13 minutes, 12 seconds

It takes up the oxygen and then it gives up carbon dioxide.

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13 minutes, 17 seconds

So it becomes de-oxygenated and then it flows back in and to the inferior vena cava, into the vein. I want to make it clear-- it's a loop.

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13 minutes, 26 seconds

It's a big loop.

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13 minutes, 27 seconds

The blood doesn't just flow out someplace and then come back someplace else. This is just one big loop.

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13 minutes, 33 seconds

And if you want to know at any given point in time, depending on your size, there's about five liters of blood.

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13 minutes, 42 seconds

And I looked it up-- it takes the average red blood cell to go from one point in the circulatory system and go through the whole system and come back, 20 seconds.

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13 minutes, 52 seconds

That's an average because you can imagine there might be some red blood cells that get stuck someplace and take a little bit more time and some go through the completely perfect route.

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13 minutes, 59 seconds

Actually, the 20 seconds might be closer to the perfect route. I've never timed it myself.

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14 minutes, 4 seconds

But it's an interesting thing to look at and to think about what's connected to what.

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14 minutes, 7 seconds

You have these these arteries up here that they first branch off the arteries up here from the aorta into the head and

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14 minutes, 17 seconds

the neck and the arm arteries and then later they go down and they flow blood to the rest of the body.

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14 minutes, 26 seconds

So anyway, this is a pretty interesting idea.

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In the next video, what I want to do is talk about, how does the hemoglobin know when to dump the oxygen?

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14 minutes, 40 seconds

Or even better, where to dump the oxygen-- because maybe I'm running so I need a lot of oxygen in the capillaries around my thigh muscles.

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14 minutes, 47 seconds

I don't need them necessarily in my hands.

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14 minutes, 49 seconds

How does the body optimize where the oxygen is actually delivering? It's actually fascinating.